Polar Ensemble Decoding of Binary Linear Block Codes

Pin-Jing Li
Advised by Prof. Yu-Chih Huang
Institute of Communications Engineering, National Yang Ming Chiao Tung University
{ouo.ee11@nycu.edu.tw, jerryhuang@nycu.edu.tw}

Abstract

Polar transformations provide a unified approach for decoding general binary linear block codes using polar-style decoders. The original polar decoding framework, PD, embeds power-of-two-length BLBCs into polar subcodes with dynamic frozen constraints. Its extension, PD+, incorporates shortening and pruning to support arbitrary blocklengths and improve transformation design. Both frameworks, however, rely primarily on sequential SC or SCL decoding.
Building on this line of work, we introduce polar ensemble decoding (PED), which enables parallel decoding by applying code automorphisms to the transformation parameter. We show that automorphisms induce classes of equivalent polar transformations with identical dynamic frozen constraints, allowing all ensemble members to share one decoder design.

TL;DR: Toward Universal Polar Decoding of Linear Codes

Polar Decoding (**PD**): Embed general BLBCs into polar subcodes through parameterized transformations.
Enhanced Polar Decoding (**PD+**): Extend the framework to arbitrary lengths using shortening and pruning.
Polar Ensemble Decoding (**PED**): Exploit automorphism-equivalent transformations for parallel decoding.

Key identity: Automorphisms of the BLBC leave the dynamic frozen constraint unchanged.

Introduction

- **Motivation.** No single BLBC family optimizes decoding complexity, minimum distance, and asymptotic performance simultaneously [1]. Standards like 5G NR mix families (LDPC for data, polar for control) at the cost of multiple dedicated decoders, motivating a **universal decoder** for multiple code families.
- **Prior art.** The parameterized universal BLBC decoding framework [2] embeds arbitrary BLBCs into polar subcodes with dynamic frozen constraints [3], later extended to arbitrary lengths [4], U-UV codes [5], and non-binary codes [6]. All rely on SCL decoding and remain *inherently sequential*; parallelization is unexplored.
- **Automorphism ensemble decoding (AED)** parallelizes soft-decision decoding for specific families. Reed-Muller [7], polar [8, 9, 10], eBCH [11], while classical automorphism use (e.g., MacWilliams' permutation decoding [12]) is confined to hard-decision error shifting. Existing schemes thus remain *code-family-dependent*.
- **Our approach.** We propose *polar ensemble decoding* (PED): applying the automorphism group to the **transformation parameter** rather than the codeword space yields equivalent polar transformations with a **shared subcode structure** yet decoding diversity, enabling permutation parallelism.

Background: Polar Code

Notation: $\mathcal{N}(\cdot)/\mathcal{R}(\cdot)$ denote the nullspace/rowspace of a matrix, RREF$(\cdot)$ its unique row-reduced echelon form, and Sym(n) the symmetric group on n elements.

Conventional Polar Codes [13]. A polar code of blocklength $n = 2^m$ has generator matrix $\mathbf{G}_p = \mathbf{B}_m\mathbf{F}^{\otimes m}$, where $\mathbf{F} = \left[\begin{smallmatrix}1 & 0\\ 1 & 1\end{smallmatrix}\right]$ is the Arıkan kernel and $\mathbf{B}_m$ a bit-reversal permutation, recursively polarizing identical channels into subchannels of varying reliability. A k-bit message $\mathbf{m}$ fills the k most reliable channels $\mathcal{I}$ ($\mathbf{u}_{\mathcal{I}} = \mathbf{m}$, $\mathbf{u}_{\mathcal{I}^c} = \mathbf{0}$), giving codeword $\mathbf{c} = \mathbf{u}\mathbf{G}_p$. SC decoding [13] estimates each u_i sequentially from $\mathbf{y}$ and $\hat{u}_1^{i-1}$; SCL [14] tracks the L most probable paths toward ML performance.

Polar Subcode [3]. Since $\mathbf{G}_p$ is invertible, any $\mathbf{c} = \mathbf{u}\mathbf{G}_p$ belongs to a BLBC $\mathcal{C}_b$ with generator $\mathbf{G}_b$ and parity-check $\mathbf{H}_b$ iff

$$\mathbf{c} \in \mathcal{N}(\mathbf{H}_b^T) \iff \mathbf{u} \in \mathcal{R}(\mathbf{G}_b\mathbf{G}_p^{-1}) =: \mathcal{R}(\mathbf{W}). \tag{1}$$

i.e., a polar subcode with dynamic frozen constraint $\mathbf{W}$, making **any BLBC polar-decoder compatible**. But the **information set is fixed by the BLBC**, with no design freedom, and may sit on unreliable channels.

Universal Polar Decoding of BLBCs

Lin *et al.* [2] added dynamic frozen-bit design flexibility via invertible matrices $\mathbf{G}_p$ and permutations.

Proposition 1. *For an $(n = 2^m, k)$ BLBC $\mathcal{C}_b$ and a permutation matrix $\mathbf{P}$, there exists a polar subcode $\mathcal{C}_p$ with dynamic frozen-bits such that the mapping between $\mathbf{c}_b \in \mathcal{C}_b$ and $\mathbf{c}_p \in \mathcal{C}_p$ is given by $\mathbf{c}_b = \mathbf{c}_p\mathbf{P}^{-1}$.*

Proof.

- **Define encoding message space**: Consider $\mathbf{u} \in \mathcal{R}(\mathbf{W_P})$, where $\mathbf{W_P} := \mathbf{G}_b\mathbf{P}\mathbf{G}_p^{-1}$
- **Rewrite BLBCs with polar basis**: Any $\mathbf{c}_b \in \mathcal{C}_b$ is $\mathbf{c}_b = \mathbf{m}\mathbf{G}_b = \mathbf{m}\mathbf{W_P}\mathbf{G}_p\mathbf{P}^{-1}$.
- **Dynamic Frozen Constraints**: Reducing $\mathbf{W_P}$ to RREF via $\mathbf{E} \in \mathrm{GL}_k(\mathbb{F}_2)$ gives $\mathbf{M_P} = \mathbf{E}\mathbf{W_P}$, so $\mathbf{c}_b = (\mathbf{m}\mathbf{E}^{-1}\mathbf{M_P}\mathbf{G}_p)\mathbf{P}^{-1} =: \mathbf{c}_p\mathbf{P}^{-1}$.
- **Reinterpretation**: $\mathbf{M_P}$ is a dynamic frozen matrix encoding message $\mathbf{m}_p := \mathbf{m}\mathbf{E}^{-1}$ into a polar subcode spanned by $\mathbf{M_P}\mathbf{G}_p$.

Polar decoders now handle general BLBCs with frozen-bit design flexibility.

Extension to Arbitrary Block Lengths

To further apply this framework to binary linear codes with arbitrary length, Lin *et al.* [4] introduced the shortened polar kernel as the base code, and a pruning operation (disabling edges of the polar kernel) that achieves a better Bhattacharyya sum optimization result.

Proposition 2. *Consider an (n, k) BLBC $\mathcal{C}_b$ and a $\mathbf{P} \in$ Sym(N), and consider an integer $N = 2^m > n$. There exists a pruned and shortend polar subcode $\mathcal{C}_p$ which was generated by a pruned polar kernel $\widetilde{\mathbf{G}}_p \in \mathbb{F}_2^{N\times N}$ and a shortening matrix $\mathbf{S} \in \mathbb{F}_2^{N\times n}$*

$$\mathcal{C}_p = \mathcal{R}(\widetilde{\mathbf{G}}_p\mathbf{S})$$

with dynamic frozen-bits such that the mapping between $\mathbf{c}_b \in \mathcal{C}_b$ and $\mathbf{c}_p \in \mathcal{C}_p$ is given by $\mathbf{c}_b = \mathbf{c}_p\mathbf{P}^{-1}$.

Parallelism of Polar Decoding

A natural parallel strategy is to decode multiple transformations simultaneously, but this is unexplored since a large set of suitable parameters is hard to obtain. We characterize the polar transformation structure to derive one.

Algebraic Characterization. Let $\pi \in$ Sym(n) act on a codeword/matrix by right multiplication with its permutation matrix $\mathbf{P}_\pi$. The polar transformation with parameter π maps $\mathcal{C}_b \to \mathcal{C}_p$, with dynamic frozen constraint $\mathbf{W}_\pi = \pi(\mathbf{G}_b)\mathbf{G}_p^{-1}$, giving the algebraic form of Proposition 1, by taking $\mathbf{m} \in \mathbb{F}_2^k$, $\mathbf{u} = \mathbf{m}\mathbf{W}_\pi$ and thus $\mathbf{u}\mathbf{G}_p = \mathbf{m}\mathbf{G}_b\mathbf{P}_\pi = \pi(\mathbf{c}_b)$.

Definition 3. (*Dynamic Frozen Constraint.*) The mapping ϕ : Sym$(n) \to \mathbb{F}_2^{k\times n}$ given by

$$\phi(\pi) = \mathbf{M}_\pi := \mathrm{RREF}(\pi(\mathbf{G}_b)\mathbf{G}_p^{-1})$$

preserves the rowspace of $\mathbf{W}_\pi$ in SC-compatible form.

- No closed-form relates π to $\phi(\pi)$, so optimizing over $|\mathrm{Sym}(n)| = n!$ is hard.
- [2] specified eBCH constructions, and [4] explored an AI-assisted search. Both find only a *single* parameter.
- Existing AED techniques [15, 10] need static-frozen information-set designs incompatible with $\mathbf{M_P}$.
- Instead, we study ϕ's **equivalence classes**, a set of parameters sharing the same decoder design.

Equivalence Classes of Polar Transformations

In the following we denote the automorphism of a code as $\mathrm{Aut}(\mathcal{C}) = \{a \in \mathrm{Sym}(n) : a(\mathcal{C}) = \mathcal{C}\}$.

Theorem 4. *(Invariance of ϕ.) For any $a \in \mathrm{Aut}(\mathcal{C}_b)$,*

$$\boxed{\phi(\pi \circ a) = \phi(\pi)}.$$

Proof. $a \in \mathrm{Aut}(\mathcal{C}_b)$ gives $\mathbf{G}_b\mathbf{P}_a = \mathbf{E}\mathbf{G}_b$ for some $\mathbf{E} \in GL_k(\mathbb{F}_2)$, so $\mathbf{G}_b\mathbf{P}_{(\pi\circ a)} = \mathbf{E}(\mathbf{G}_b\mathbf{P}_\pi)$ and hence $\mathcal{R}(\mathbf{G}_b\mathbf{P}_{(\pi\circ a)}\mathbf{G}_p^{-1}) = \mathcal{R}(\mathbf{G}_b\mathbf{P}_\pi\mathbf{G}_p^{-1})$. Since RREF is canonical for rowspace, $\phi(\pi \circ a) = \phi(\pi)$.

Remark 1. The coset $\pi\mathrm{Aut}(\mathcal{C}_b)$ is generally neither Aut$(\mathcal{C}_b)$ nor the polar subcode's automorphism group. Using a itself as π collapses to Trifonov's fixed frozen set since $\phi(a) = \phi(\mathrm{id})$.

Corollary 5. *(Polar Transformation Equivalent Class.) Every $\sigma \in \pi Aut(\mathcal{C}_b)$ yields the same image polar subcode $\mathcal{C}_p = \{\mathbf{c}_p = \mathbf{u}\mathbf{G}_p : \mathbf{u} \in rowspan(\phi(\pi))\}$. Identical dynamic frozen constraints induce equivalent transformations.*

Remark 2. Theorem 4 shifts the automorphism action from the codeword space to the **parameter** π, unlike prior AED methods, so well-characterized BLBC automorphism groups directly furnish equivalent transformations without code-specific redesign.

Example 1. *Repetition Code.* Consider a repetition code with generator matrix $\mathbf{G} = \left[\begin{smallmatrix}1&1&1&0&0&0&0&0\\ 0&0&0&1&1&1&0&0\\ 0&0&0&0&0&0&1&1\end{smallmatrix}\right]$, built from a length-3, a length-3, and a length-2 repetition block. Permuting within a block, or exchanging the two length-3 blocks, is an automorphism. For $\mathbf{P}_1 = (1\,4)(2\,5)(3\,6)$ (block exchange), $\mathbf{G}\mathbf{P}_1^{-1}$ swaps the first two block-rows, so RREF gives $\mathbf{M}_{\mathbf{P}_1} = \mathbf{G}\mathbf{G}_n^{-1}$; for $\mathbf{P}_2 = (1\,2)$ (within-block swap), $\mathbf{G}\mathbf{P}_2^{-1} = \mathbf{G}$ directly, giving $\mathbf{M}_{\mathbf{P}_2} = \mathbf{G}\mathbf{G}_n^{-1}$ too. Two structurally different automorphisms thus induce the *same* dynamic frozen matrix, confirming Theorem 4.

Proposed Algorithm

Algorithm 1: Polar Ensemble Decoding for BLBCs

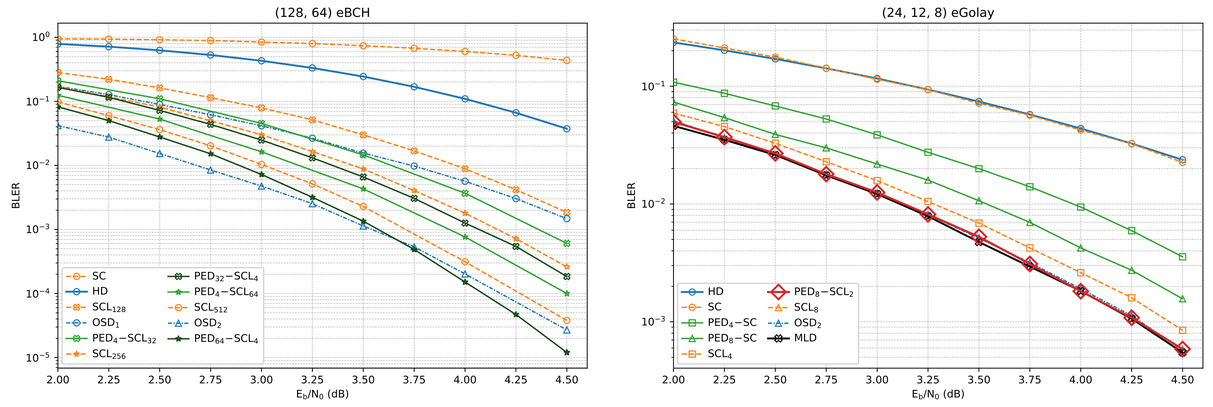
Input: Channel output $\mathbf{y}$, base permutation π, parallelization path size $M \le |\mathrm{Aut}(\mathcal{C}_b)|$, automorphism subset $\{a_1, \ldots, a_M\} \subseteq \mathrm{Aut}(\mathcal{C}_b)$, RREF Elimination matrices $\{\mathbf{E}_1, \ldots, \mathbf{E}_M\}$

for $i = 1$ **to** M **do**
  $\mathbf{y}_i \leftarrow (\pi \circ a_i(\mathbf{y}))$;
  $\hat{\mathbf{c}}_i \leftarrow \mathrm{PolarDec}(\mathbf{y}_i)$;
$j \leftarrow$ **Combiner**$(\{\hat{\mathbf{c}}_i\}_{i=1}^M)$;
$\hat{\mathbf{m}} \leftarrow \hat{\mathbf{m}}_j\mathbf{E}_j$;

Output: Estimated message $\hat{\mathbf{m}}$

Algorithm 1 (depicted in the figure at the bottom of the poster) summarizes PED: all M parallel decoders admit the same decoder structure since $\phi(\pi \circ a) = \phi(\pi)$, and candidates are combined by (i) best polar-decoder path metric, or (ii) BLBC parity-check validity. Varying M trades off SCL-equivalent latency, complexity, and hardware cost at a fixed effective list size.

Simulation Results



(a) (128, 64) eBCH code. (b) (24, 12) eGolay code.

Figure: BLER performance of PED over the AWGN channel, against ML, SC, SCL, and HD decoding.

PED_M denotes M permutations from the automorphism coset of base parameter π [2], followed by SC or SCL_L.

- **(128,64) eBCH (Fig. 1a).** At equal effective list size, PED_{64}-SCL_4 outperforms PED_4-SCL_{64} by ∼0.5 dB and beats SCL_{512} using only half the list size. Each PED path explores a distinct region via a different automorphism, whereas larger SCL lists only refine a single search trajectory.
- **(24,12) eGolay (Fig. 1b).** Generality beyond power-of-two lengths (automorphism group M_{24} [16], multi-kernel $\mathbf{G}_p$ [17]): PED improves substantially over plain SC and nearly matches SCL at equal list size, with $M = 8$, $L = 2$ approaching ML and comparable to $\mathrm{SCL}_{64}/\mathrm{OSD}_2$ at lower latency.
- **Takeaway.** PED achieves flexible latency-parallelism tradeoffs across BLBC families with nontrivial automorphisms by operating on transformation parameters rather than codewords.

Conclusion

We proposed PED, a parallelization of the parameterized universal polar decoding framework for arbitrary BLBCs with nontrivial automorphism. By exploiting the automorphism structure of the embedded BLBC, decoding within the automorphism right coset induces equivalent polar subcodes that are decoded independently using standard polar-style decoders. PED thus provides scalable tradeoffs among decoding performance, latency, and implementation complexity for parallel soft-decoding of BLBCs.

Future Work

- **Decoding Architecture**: Diversity analysis, Combiner Design, Base polar code design, Non-binary code
- **Polar Transformation**: Effective transformation parameter search, Algebraic Properties
- **Relations to Other Topics**: Trellis optimization, Code-base cryptography

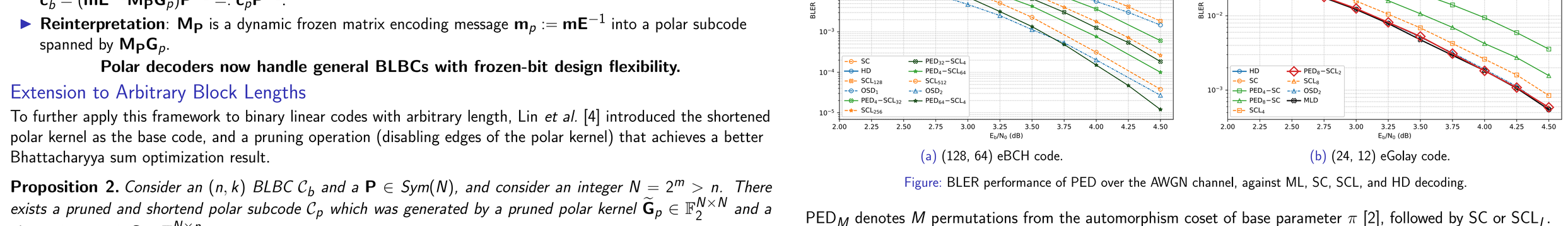


Figure: Overall PED architecture. All the dashed blocks are virtual components.

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